

CARMACKS, YT, Canada

WMO: 710390

Lat: 62.1150N

Lon: 136.1919W

Elev: 543

StdP: 94.97

Time Zone: -8.00 (NAP)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.8	-40.0	-46.6	0.0	-41.9	-43.5	0.1	-38.8	8.2	0.2	7.1	-1.1	0.1	130	0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	27.0	14.0	25.0	13.3	23.2	12.8	15.7	23.4	14.9	22.3	14.1	20.8	3.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	10.1	16.5	12.1	9.4	16.0	11.0	8.7	15.4	45.5	23.6	43.3	22.4	41.1	20.8	17.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.3	6.3	5.4	-44.4	30.4	2.1	2.2	-45.9	31.9	-47.1	33.2	-48.3	34.4	-49.9	36.0
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.1	-21.1	-17.3	-9.7	0.6	8.0	13.4	15.1	12.6	6.9	-1.0	-13.8	-20.0	(6)	
(7)		DBStd	14.71	10.71	9.57	8.53	5.00	3.77	3.06	2.41	3.12	3.60	5.17	8.52	9.65	(7)	
(8)		HDD10.0	4809	964	763	612	281	84	4	1	10	104	342	714	929	(8)	
(9)		HDD18.3	7468	1222	997	870	531	319	152	103	178	343	600	964	1188	(9)	
(10)		CDD10.0	390	0	0	0	0	23	106	158	91	11	1	0	0	(10)	
(11)		CDD18.3	7	0	0	0	0	0	3	3	1	0	0	0	0	(11)	
(12)		CDH23.3	396	0	0	0	0	26	142	137	91	1	0	0	0	(12)	
(13)		CDH26.7	81	0	0	0	0	2	41	21	18	0	0	0	0	(13)	
(14)	Wind	WSAvg	1.8	1.0	1.1	1.9	2.2	2.5	2.3	2.2	2.1	2.1	1.7	1.3	0.9	(14)	
(15)	Precipitation	PrecAvg	252	12	7	4	4	22	36	58	40	30	17	11	9	(15)	
(16)		PrecMax	323	29	21	14	11	46	80	107	73	65	40	30	21	(16)	
(17)		PrecMin	172	0	1	0	0	0	8	26	7	0	3	1	2	(17)	
(18)		PrecStd	45	8	6	4	3	12	18	23	17	16	8	6	5	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	5.6	11.5	19.1	26.0	31.0	29.1	29.0	21.2	14.8	6.5	4.4	(19)	
(20)			MCWB	3.1	1.9	4.9	11.7	11.7	15.5	15.5	15.2	11.6	8.7	3.1	1.5	(20)	
(21)		2%	DB	1.7	2.9	7.4	14.4	23.5	27.0	26.4	26.0	18.3	11.1	2.8	1.5	(21)	
(22)			MCWB	-0.2	0.4	2.0	5.8	10.9	13.3	14.7	14.4	10.8	6.5	0.7	-0.3	(22)	
(23)		5%	DB	-1.0	0.2	5.2	11.6	20.7	24.3	24.7	23.2	16.5	8.8	-0.2	-1.8	(23)	
(24)			MCWB	-2.2	-2.0	0.6	4.1	9.5	12.4	13.8	13.5	9.7	4.6	-1.1	-2.7	(24)	
(25)		10%	DB	-5.6	-2.8	3.0	9.7	18.2	22.2	23.0	20.7	14.7	6.5	-2.9	-6.4	(25)	
(26)			MCWB	-6.0	-4.2	-0.7	3.1	8.6	12.1	13.4	12.5	8.8	3.7	-3.4	-6.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	2.6	6.2	7.9	12.9	15.8	16.4	16.3	13.3	7.9	2.1	1.8	(27)	
(28)			MCDWB	5.5	5.6	13.3	14.4	25.0	26.7	23.2	25.1	19.7	11.9	4.2	3.8	(28)	
(29)		2%	WB	0.6	0.3	4.1	5.6	11.8	14.3	15.8	15.3	11.7	6.2	0.3	0.0	(29)	
(30)			MCDWB	1.7	3.2	9.8	12.0	22.8	23.6	22.7	24.3	18.1	9.1	1.9	2.0	(30)	
(31)		5%	WB	-1.2	-2.4	2.2	4.7	10.8	13.3	15.3	14.4	10.4	5.1	-1.4	-2.4	(31)	
(32)			MCDWB	0.3	-0.7	6.8	10.8	20.9	21.7	21.9	22.3	16.4	8.0	-0.5	-1.2	(32)	
(33)		10%	WB	-3.6	-5.2	0.6	3.8	9.7	12.4	14.5	13.5	9.5	3.7	-3.5	-6.3	(33)	
(34)			MCDWB	-2.8	-4.0	4.9	9.6	18.5	20.3	20.7	20.2	14.6	6.1	-2.8	-5.9	(34)	
(35)	Mean Daily Temperature Range			MDBR	9.5	13.7	16.4	14.8	15.9	15.1	14.0	13.7	13.1	9.6	8.4	8.8	(35)
(36)		5% DB	MCDBR	13.5	14.3	15.5	17.4	21.5	19.7	18.3	19.2	16.1	12.7	11.1	13.2	(36)	
(37)			MCWBR	12.0	10.8	10.3	9.4	10.5	7.9	6.8	7.9	8.9	8.2	9.7	9.8	(37)	
(38)		5% WB	MCDWB	13.5	13.5	14.7	14.8	20.8	15.9	13.8	16.1	14.9	10.8	11.6	11.1	(38)	
(39)			MCWBR	11.8	10.8	8.7	8.7	10.1	6.9	6.0	7.6	8.5	7.4	10.0	9.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.224	0.249	0.260	0.343	0.367	0.372	0.372	0.367	0.303	0.273	0.205	0.173	(40)	
(41)		taud		2.005	2.247	2.288	2.126	2.327	2.352	2.371	2.399	2.560	2.481	2.070	1.932	(41)	
(42)		Ebn at Noon		447	736	868	835	850	851	840	812	822	715	486	333	(42)	
(43)		Edh at Noon		37	60	83	123	110	110	105	94	66	49	32	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg		0.30	1.14	2.90	4.83	5.48	5.67	5.08	4.04	2.58	1.18	0.39	0.14	(44)	
(45)		RadStd		0.02	0.07	0.15	0.24	0.25	0.33	0.28	0.32	0.23	0.13	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	N/A
(47) Regional (5 neighbors)		N/A	+1.87	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page